

Section 4.4: Final Exam Review

Finite Mathematics MTH 107 Fall 2014

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Mars Hill University

Friday 14th November, 2014



Schedule

<http://www.mhu.edu/academics/course-catalog/exam-schedule>

Final Exam	Date	Time
Section 07	Friday 5 th December	8-10am
Section 10	Wednesday 3 rd December	2:30-4:30pm

Chapter 1: Logic

- Words into symbols
- Symbols into words

Chapter 1: Logic

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Example 4.4.1.

We do our homework or we do not miss class.

- Symbols into words

Chapter 1: Logic

- Words into symbols

Example 4.4.1.

We do our homework or we do not miss class.

- Symbols into words

Example 4.4.2.

Let h represent "We do our homework", and m represent we miss class". Symbolize: *If we miss class then we did not do our homework.*

Chapter 1: Logic

- Negation
 - All $\leftrightarrow$ some are not
 - None $\leftrightarrow$ some are
- Construct a truth table

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 - All $\leftrightarrow$ some are not
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Example 4.4.3.

Negate: Some people are not going to have fun

- Construct a truth table

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 - All $\leftrightarrow$ some are not
 - None $\leftrightarrow$ some are

Example 4.4.3.

Negate: Some people are not going to have fun

- Construct a truth table

Example 4.4.4.

Construct a truth table for $(p \vee q) \rightarrow \sim q$

Chapter 1: Logic

- Use a truth table to determine if an argument is valid.

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- Use a truth table to determine if an argument is valid.

Example 4.4.5.

m : You like money. 1. If you like food then you like money.
 f : You like food. 2. You do not like food.

Therefore, you do not like money.

		1	2	$1 \wedge 2$	c	Argument
f	m					
T	T					
T	F					
F	T					
F	F					

Chapter 2: Sets

- Basic set operations

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Example 4.4.6.

Let $A = \{a, b, c, d\}$, $B = \{c, d, e, f\}$, $C = \{e, f, g\}$, and $U = \{a, b, c, d, e, f, g, \}$.

- Find $A \cup B$
- Find $A \cap B$
- Find $A \cap C$
- Find A'
- Find $(A \cup B)'$
- Find $A' \cap B'$
- Find $n(A)$
- Find $n(A \cup B)$

Chapter 2: Sets

- Venn Diagrams

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- Venn Diagrams

Example 4.4.7.

A class has 50 students. 20 are biology majors, 30 are math majors, and 10 are math and biology majors. Construct a Venn diagram and label the number of students in each region

Chapter 2: Counting

- Basic operations:

- Basic operations:

Example 4.4.8.

Find the following:

- $5!$
- $\frac{100!}{98!}$
- ${}_5P_3$
- ${}_5C_3$